

PSI FIELD LAB

Synthetic Calibration and Prospective Field Validation Protocol

A data-centered methodology for documenting, synchronizing, auditing, and reviewing multimodal field observations in anomalous and paranormal research contexts

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Core boundary statement: this paper proposes a documentation, calibration, and field-validation methodology. It is not designed to prove or disprove paranormal causation.

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Intellectual Priority and Versioning Statement

This document establishes the methodological framework, terminology, staged validation structure, analytical workflow, and governance assumptions of the PSI Field Lab as developed by Alberto Cardenas and Paranormal 3:33 Research Lab. Future work that uses, adapts, implements, or extends this framework should cite this version or the corresponding repository version of record. The purpose of this statement is not to restrict independent research into anomalous or paranormal claims, but to preserve a clear record of this specific data-centered methodology and its authorship.

Abstract

This paper presents the PSI Field Lab as a two-stage research methodology for documenting, synchronizing, auditing, and reviewing multimodal field observations in anomalous and paranormal research contexts. Stage I is a synthetic calibration demonstration. Its records were generated computationally with a fixed seed and a documented experimental structure designed to resemble the derived data that a future field study could produce. It tests data integration, quality gates, transparent scoring, unsupervised anomaly detection, site-level repeatability, sensitivity analysis, constrained language-model review, and run-level audit packaging. It does not contain real sensor recordings or evidence about paranormal phenomena.

Stage II is a prospective real-data protocol. It specifies how physiological, EEG, motion, video, audio, environmental, and contextual records would be collected using a Muse 2 headset, a Polar heart-rate system capable of exporting beat-to-beat intervals, body cameras, an independent audio recorder, and environmental instruments. It defines clock synchronization, immutable raw-data retention, transformation into a common ten-second analytical schema, matched target and control sessions, randomized order, blinded review, quality gates, exclusions, and repetition requirements.

Across both stages, the system is restricted to documentation and evidence triage. A high score means unusual under recorded conditions, not paranormal. The methodological contribution is a traceable bridge from synthetic software calibration to preregistered field acquisition, with observation, interpretation, explanation, and causal claims kept separate.

Keywords: paranormal research; parapsychology; synthetic calibration; field validation; multimodal sensors; Muse 2; heart-rate variability; anomaly triage; preregistration; repeatability; language-model review; research ethics.

1. Research Position and Scope

Paranormal and parapsychological field research is often pulled toward two opposed errors. One begins with belief and interprets every unusual observation as evidence. The other begins with rejection and treats every unusual observation as error. The PSI Field Lab takes a third position: the unknown deserves disciplined documentation before interpretation. An event can be unusual, unresolved, or worthy of follow-up without becoming evidence of spirits, survival of consciousness, telepathy, or any other extraordinary cause.

The methodology separates four levels that are routinely mixed together:

- Observation: what a record states, such as a timestamp, sensor value, audio transient, motion score, or environmental condition.
- Interpretation: a provisional human reading of that record.
- Explanation: a conventional or unresolved account that may fit the evidence.
- Claim: a stronger statement about cause.

The PSI Field Lab is limited to observations, interpretations, and review priorities. No rule, statistic, machine-learning model, or language-model reviewer described here is authorized to make a paranormal causal claim.

1.1 Why This Version Exists

Version 1.0 formalizes the paper for public citation and archival posting. Earlier working versions were prepared for methodological discussion. This version adds authorship, contribution, citation, license, intellectual-priority, and platform-readiness language while preserving the core technical distinction between synthetic calibration and prospective empirical validation.

2. Original Contribution

The paper contributes a specific framework rather than a claim of discovery. Its original contribution is the combination of:

- a two-stage sequence from synthetic software calibration to prospective real-data field validation;
- explicit separation of observation, interpretation, explanation, and causal claim;
- multimodal sensor alignment using ten-second analytical windows and immutable raw archives;
- gate-first scoring that treats quality, motion, and conventional explanations as eligibility conditions;
- participant-level site repeatability rather than raw event counts;
- constrained language-model review as an audit aid rather than an oracle;
- run-level packaging with manifests, hashes, configuration, prompts, model identity, and visible limitations;
- a governance model for preregistration, consent, participant wellbeing, and controlled data release.

3. Stage I - Synthetic Calibration Demonstration

3.1 Purpose of the Demonstration

The current PSI Field Lab notebook operates on a synthetic calibration bench. The bench exists to test the analytical instrument under conditions whose answers are known in advance. It asks whether the pipeline can elevate designed multimodal candidates, demote known movement and environmental artifacts, distinguish cross-participant site repetition from repeated activity in one participant, expose poor model performance, and preserve every result in a run-specific package.

The demonstration is not a reconstruction of a completed investigation. No participant wore a headset or heart-rate monitor; no camera or audio recorder captured a location; no environmental probe was read in the field. Instead, the generator directly produces the derived variables that a real acquisition-and-transformation pipeline is expected to produce.

3.2 Synthetic Data Provenance

The dataset is generated by *generate_psi_data_v2.py* with random seed 333. The generator creates a structured experimental universe and writes five canonical comma-separated files.

File	Rows	Role
sessions.csv	162	Protocol and contextual metadata for primary and repeated sessions.
sensor_windows.csv	28,122	Ten-second windows containing simulated physiological, EEG-style, motion, audio, environmental, and signal-quality features.
events.csv	220	Timestamped candidates with event-level deltas computed from the preceding local baseline.
audio_reviews.csv	160	Simulated review outcomes for designed audio evidence plus a sample of control events.
ground_truth.csv	220	Generator-only labels identifying designed positives, hard negatives, and site clusters. Loaded only after analysis.

3.3 Intended Sensor Correspondence

Feature family	Intended real source	Required transformation
Heart rate, RR, RMSSD, SDNN	Polar heart-rate system with beat-to-beat interval export.	Preserve raw RR intervals, flag invalid beats under frozen rules, and compute windowed heart-rate and HRV features.
Relative EEG-style bands and motion	Muse 2 EEG and accelerometer streams.	Retain raw channels, document contact quality, reject motion-contaminated periods, estimate spectral power, and normalize within windows.
Motion and light-change scores	Timestamped body-camera video.	Preserve original video, compute frame-level motion and luminance change, and aggregate to the common window.
Audio RMS, peaks, transients	Independent timestamped audio recorder.	Preserve source audio, document gain and sample rate, extract acoustic features, and retain clips for blinded review.
Environmental and contextual covariates	Environmental instruments and structured field log.	Align device logs and manual observations to the common clock while recording known conventional sources.

3.4 What Stage I Can and Cannot Establish

Stage I can test loading, joins, event deltas, session baselines, quality gates, ranking behavior, participant-level site repetition, sensitivity reporting, constrained language-model review, and run packaging. It cannot validate sensor accuracy, synchronization under real clock drift, electrode contact, RR artifact correction, camera occlusion, audio contamination, investigator behavior, participant expectation, site effects, or any paranormal hypothesis.

Protocol rule: *all demonstration results must be labeled synthetic in the notebook, paper, tables, figures, and exported summaries. No synthetic performance metric may be presented as field evidence.*

4. Stage II - Prospective Real-Data Field Protocol

4.1 Research Objective

The real-data phase asks a bounded empirical question: under prospectively fixed rules, do predefined multimodal candidate patterns occur more consistently in target-site sessions than in matched control sessions, and do they replicate across independent participants and collection waves? This formulation does not assume a paranormal cause. It tests whether an observable, quality-controlled pattern repeats after conventional explanations, movement, device failure, and environmental context are considered.

4.2 Instruments and Raw Records

- Muse 2 headset for raw EEG channels, contact-quality information, and accelerometry.
- Polar heart-rate system capable of exporting beat-to-beat RR intervals, with exact model and firmware fixed before enrollment.
- One or more body cameras with original timestamped video retained.
- Independent audio recorder producing a lossless or minimally compressed source file.
- Environmental instruments for temperature, humidity, ambient sound level, and any additional physical channel selected before preregistration.
- Structured field log documenting location, zone, investigator actions, known animals, occupancy, electrical infrastructure, running water, weather, traffic, and deviations.

The exact camera, audio-recorder, environmental-sensor, and Polar models must be recorded in a device registry before real collection. The registry should include model, serial or asset identifier, firmware, sampling rate, clock behavior, placement, calibration, software version, and export format.

4.3 From Physical Signals to Analytical Dataset

The real pipeline begins with immutable raw recordings. Derived CSV files are outputs of a documented transformation, not substitutes for the original evidence. Each native file is copied into a read-only raw archive, hashed, converted to a common UTC timeline while retaining original device timestamps, corrected for clock offset and drift, transformed under frozen preprocessing rules, and aggregated into non-overlapping ten-second windows with explicit missingness and quality flags.

4.4 Standard Field Session

A complete field session includes a five-minute baseline, a twenty-minute active period, and a five-minute recovery period. Target and control sessions should use equivalent activity scripts, investigator behavior, timing, sensor placement, and environmental logging. Condition order should be randomized or counterbalanced where possible.

4.5 Target and Control Conditions

A target location is a site selected before data collection because of historical, testimonial, environmental, or investigative relevance. A matched control location should resemble the target in structure, access, acoustics, lighting, safety, investigator movement constraints, and ordinary environmental sources. The control is not an empty place; it is a structured comparison condition.

4.6 Repetition Framework

Each participant completes two valid sessions in each condition on separate days. An initial feasibility study uses twelve independent participants per target-control pair. Any confirmatory claim requires a new participant wave with sample size fixed by an a priori power analysis rather than by an arbitrary repetition count.

4.7 Session Acceptance and Exclusion

A session is valid for primary analysis only if preregistered requirements are satisfied: valid consent, completed phases, required sensor streams, recoverable synchronization markers, acceptable residual alignment error, sufficient usable windows, documented protocol adherence, and recorded physiological covariates. Event-level

eligibility remains separate from session-level validity. A valid session can contain an ineligible event, and one clean event cannot rescue an invalid session for primary analysis.

4.8 Primary Outcome and Confirmatory Decision

Before real enrollment, the protocol must select one primary outcome. A defensible choice is the participant-level proportion who produce at least one strong, gate-passing multimodal candidate in the target condition compared with the matched control condition. The participant, not the event, is the primary experimental unit for site-level inference.

A site result is methodologically supported only when the preregistered target-control contrast passes its inferential criterion in the confirmatory wave, the direction is consistent across collection sessions, no single participant drives the result, quality and conventional-explanation gates remain satisfied, and the result is not dependent on one narrow scoring-weight choice. Even then, the conclusion is a replicated unexplained multimodal pattern under study conditions, not a paranormal cause.

5. Shared Analytical and Review Framework

5.1 Candidate Detection from Synchronized Windows

For each valid session, the participant's five-minute baseline establishes the reference for heart rate, HRV, and relative EEG-band activity. Every subsequent usable window is standardized against that participant's own baseline rather than against another participant or a fixed population threshold. Detector segments and field annotations remain separate evidence sources. Their agreement can increase review priority, but an investigator annotation cannot force an event to pass a quality gate.

5.2 Transparent Scoring: Gates Before Points

Recording quality is a precondition, not evidence of an anomaly. Eligibility gates determine whether an event can be scored: session eligibility, signal quality, low motion, and absence of a documented conventional explanation. Only eligible events receive evidence points. Site reputation and prior number of candidates are excluded from event-level scoring.

5.3 Repeatability and Corrected Site Analysis

The site analysis counts distinct participants with at least one strong candidate. It never substitutes event count for participant count. Uncorrected and multiplicity-corrected results are displayed together. A result notable before correction and unconvincing afterward remains visible as exploratory evidence, not a confirmed effect.

5.4 Unsupervised Detection and Sensitivity Analysis

An unsupervised anomaly detector provides a limited second opinion: it identifies numerical records that differ from the rest of the dataset. It cannot determine cause. The rule-based ranking is repeated under many plausible weight combinations. A candidate that appears only under one precise weighting is fragile.

5.5 Constrained Language-Model Reviewer

The language model is an evidence reviewer, not an oracle. For one candidate event it answers a fixed question: does this record deserve additional human review, and what should the reviewer check? It is never asked whether a location is haunted or what an event ultimately means. Its response must cite supplied fields, consider conventional explanations, state limitations, and remain blind to score, percentile, site rank, synthetic ground truth, and paranormal label when used in blind mode.

6. Preregistration, Governance, and Ethics

Before the pilot, the team freezes acquisition schema, quality rules, exclusions, detector logic, scoring weights, language-model prompt, primary endpoint, and planned summaries. Before confirmation, the revised protocol is preregistered with timestamp, sample-size justification, stopping rule, multiplicity plan, and distinction between confirmatory and exploratory analyses.

Physiological and EEG data are sensitive personal data. Participants receive plain-language information about each sensor, recording duration, foreseeable discomfort, data use, retention, and withdrawal. Identifiers are stored separately from analytical data, and publication files use anonymized codes and redacted notes.

The synthetic package may include complete input snapshots because it contains no real participants. Real-data packages require at least two release levels: an internal audit package for authorized research personnel and a collaborator or publication package containing approved aggregates, redacted event summaries, methods, limitations, and non-identifying metadata.

7. Run-Level Reporting and Audit Trail

Each notebook execution receives a unique UTC-based identifier and writes to a separate directory. The final notebook section presents an executive summary, principal event and site results, visible limitations, and a dynamic inventory that reopens run files and reports format, dimensions, fields, size, and validation status. The run package records input hashes, code and prompt versions, seed, data source, model identity, start and completion time, and artifact manifest.

8. Limitations and Transition Criteria

The current software has been exercised against synthetic data whose patterns were designed by the generator. Success on that bench can reveal implementation defects and analytical blind spots, but it may overestimate performance because the real world contains artifacts and dependencies the generator did not imagine.

Consumer EEG is vulnerable to motion, muscle activity, poor electrode contact, and environmental interference. HRV depends on beat detection quality and physiological context. Video and audio features can be driven by operator movement, gain control, compression, reflections, radio sources, animals, plumbing, weather, and site infrastructure. Field participants bring expectations, fatigue, medication, caffeine, prior exertion, and emotional response. These are part of the causal structure a serious protocol must record.

The project is ready to leave the synthetic-only stage only after the device registry is complete, engineering runs achieve three consecutive passes, transformation code is verified against scripted events, consent and data-governance materials are approved, pilot rules are frozen, and the target-control design is preregistered. It is ready for confirmatory language only after an independent participant wave is powered and completed under a frozen primary analysis.

9. Conclusion

The PSI Field Lab is best understood as a staged methodological instrument. Its current demonstration uses synthetic, sensor-informed derived data to test whether the analytical pipeline behaves transparently under known conditions. Its future empirical value depends on collecting real raw signals under a controlled, repeatable, and ethically governed protocol, then transforming them into the same analytical structure without changing the rules to fit the result.

The formal sequence is clear: calibrate the software on known synthetic cases; validate acquisition and synchronization through engineering runs; conduct a matched, counterbalanced feasibility pilot; preregister and power a confirmatory wave. Throughout, preserve conventional explanations, treat the participant rather than the event as the experimental unit for site inference, and describe a replicated pattern as unresolved rather than paranormal.

For Paranormal 3:33 Research Lab, the central contribution is a move from stories to structured evidence, and from a convincing demonstration to a protocol capable of being tested, audited, and repeated.

Appendix A. Synthetic Data Dictionary Summary

Table	Representative fields
Sessions	Session, site, and participant identifiers; trial and repeat linkage; date and time; duration; eligibility; device and synchronization status; building and environmental context; protocol deviations; notes.
Sensor windows	Session and window identifiers; start and end seconds; phase and zone; heart rate; RR; RMSSD; SDNN; relative EEG-style bands; accelerometry; body-camera motion and light; audio RMS, peak, and transients; environmental values; quality and artifact fields.
Events	Event, session, site, and participant identifiers; interval; type; modalities; detection source; observation; explanation; local sensor deltas; audio, motion, and environmental summaries.
Audio reviews	Review, event, session, site, and participant identifiers; review interval; recorder and spirit-box status; simulated result; intelligibility; radio and environmental-source flags; confidence; explanation.
Ground truth	Generator labels used only for synthetic post-analysis evaluation.

Appendix B. Real-Data Session Checklist

- Before collection: consent complete; participant code assigned; condition order controlled; device registry and firmware verified; storage available; clocks checked; sensor placement documented; scripted control markers recorded.
- During collection: baseline complete; activity script followed; zone and intervention markers recorded; operator movement documented; conventional sources logged; participant stop status monitored.
- After collection: closing synchronization marker recorded; device status checked; participant debriefed; native files copied to raw archive; hashes calculated; deviations recorded before analysis; target/control labels protected from blinded reviewers.
- Before primary analysis: transformation version fixed; alignment and completeness checks passed; exclusions applied without outcome access; audio review complete; code, prompts, weights, and primary endpoint frozen.

Appendix C. Derived Analytical Artifacts

Artifact	Purpose	Typical release
Event ranking	Preserves gates, evidence components, detector support, audio status, and review outcomes.	Redacted or aggregated.
Site analysis	Reports participant-level repetition, uncertainty, corrected statistics, and sensitivity.	Shareable when non-identifying.
Detector diagnostics	Preserves segments, baseline diagnostics, signal-quality behavior, and temporal relationships.	Internal by default.
Language-model audit	Retains structured summaries, consistency, blind candidates, and raw responses used for verification.	Summaries may be shared; raw records remain controlled.
Synthetic evaluation	Compares rankings against generator labels without label leakage.	Shareable for calibration only.
Run provenance	Records run identity, hashes, versions, configuration, manifest, and status.	Shareable after secret and path review.

Appendix D. Statement for Collaborators

The PSI Field Lab currently demonstrates an analytical workflow using synthetic, sensor-informed data. It does not present observations collected from real participants or physical sites. The prospective field protocol is designed to test whether predefined multimodal patterns repeat across target and matched control conditions under fixed acquisition, quality, blinding, and analysis rules. Any resulting event remains a candidate for human review, not proof of paranormal causation.

References

- Bem, D. J. (2011). Feeling the future: Experimental evidence for anomalous retroactive influences on cognition and affect. *Journal of Personality and Social Psychology*, 100(3), 407-425.
- Brysbaert, M. (2019). How many participants do we have to include in properly powered experiments? *Journal of Cognition*, 2(1), 16.
- Duane, T. D., & Behrendt, T. (1965). Extrasensory electroencephalographic induction between identical twins. *Science*, 150, 367.
- Grinberg-Zylberbaum, J., Delaflor, M., Attie, L., & Goswami, A. (1994). The Einstein-Podolsky-Rosen paradox in the brain: The transferred potential. *Physics Essays*, 7(4), 422-428.
- Grinberg-Zylberbaum, J., & Ramos, J. (1987). Patterns of interhemispheric correlation during human communication. *International Journal of Neuroscience*, 36(1-2), 41-55.
- Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, 8(1), 33267.
- Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600-2606.
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology. *Psychological Science*, 22(11), 1359-1366.
- Targ, R., & Puthoff, H. E. (1974). Information transmission under conditions of sensory shielding. *Nature*, 251, 602-607.
- Wagenmakers, E.-J., Wetzels, R., Borsboom, D., & van der Maas, H. L. J. (2011). Why psychologists must change the way they analyze their data: The case of psi. *Journal of Personality and Social Psychology*, 100(3), 426-432.
- Wiseman, R., & Watt, C. (2006). Belief in psychic ability and the misattribution hypothesis. *British Journal of Psychology*, 97(3), 323-338.